

Passive Haptic Feedback for Control Elements in Virtual Reality

Markus Tatzgern*

Salzburg University of Applied Sciences
Puch, Austria

David Fasching†

Salzburg University of Applied Sciences
Puch, Austria

Vivian Gómez‡

Universidad de los Andes
Bogotá D.C, Colombia

Sebastián Calvache§

Universidad de los Andes
Bogotá D.C, Colombia

Pablo Figueroa¶

Universidad de los Andes
Bogotá D.C, Colombia

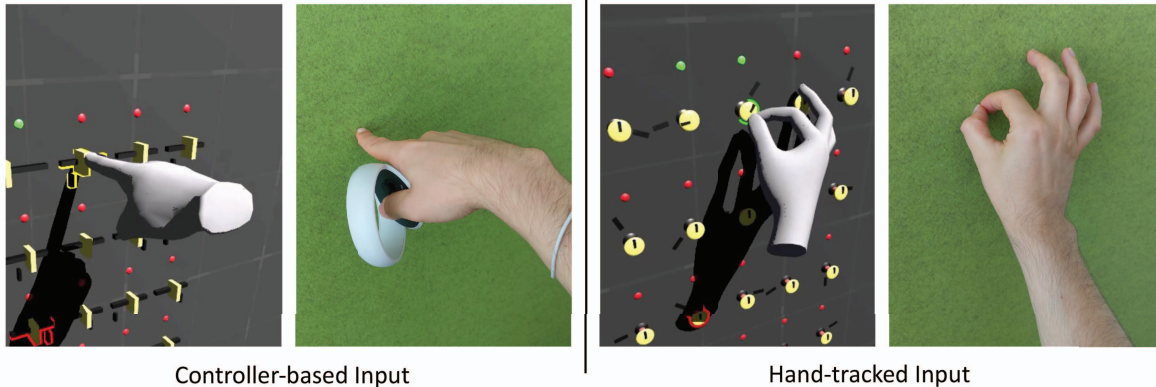


Figure 1: Haptic Feedback for Control Elements. We explore the benefit of utilizing simple passive haptic feedback when interacting with virtual control elements. (Left) Controller-based input. (Right) Hand-tracked input method. We apply both methods to typical control elements button, toggle, slider, as well as knobs.

ABSTRACT

We designed a controller-based hand input method that allows users to perceive haptic feedback with typical handheld controllers when interacting with virtual control elements while at the same time benefitting from precise controller-based tracking. We performed an experiment comparing our controller-based input method against native HMD hand tracking. Interestingly, hand tracking was robust and precise during interactions with most control elements in our experimental setup. A surprising outcome was that haptic feedback generally slowed down users during interaction, potentially due to a fear of hitting the physical surface during interaction.

1 INTRODUCTION

To increase the fidelity of a Virtual Reality (VR) simulation, physical replicas of real-world objects can be integrated into the simulation. These physical representations provide tactile and haptic feedback during interactions, which can improve the experience, as well as the performance of users [2]. While physical replicas increase the cost of simulations, research has shown that rough physical representations of the real-world could be utilized for passive haptic feedback for VR simulations [3]. We present an input approximation for hand gestures that utilizes precise VR controller-based tracking and at the same time allows users to experience tactile and haptic feedback by

touching a wall surface with the index finger of their real hand (see Fig. 1(Left)). Utilizing the index finger allows touching physical, real-world objects, while avoiding collisions between the controller and these objects. Similar to [11], we utilize the capability of the Meta Quest 2 controllers for roughly tracking finger positions on the controller and forming the approximate pointing gesture so that a user's virtual hand extends the index finger like its real counterpart.

2 RELATED WORK

Interaction performance can be improved by utilizing haptic feedback that allows users to perceive sensory feedback from touching controls. In early work, Borst et al. [2], showed that aligning a static physical surface to the virtual interaction surface to provide passive haptic feedback improves interaction performance for button and slider manipulation. Joyce et al. [6] demonstrated that selection tasks are more effective when using a static physical surface as well. Lindeman et al. [8, 9] provided a handheld physical surface that users carried in their hand and demonstrated that the performance of selection and manipulation tasks benefitted from the haptic feedback. In order to avoid additional hardware and instrumentation of the environment and increase flexibility, previous work also explored utilizing the user's body for haptic feedback during interaction, for instance, hands [7], or arms [4].

3 EXPERIMENT

We designed a within-subject study with three independent variables: control element, input method, and haptic feedback. Control element had four conditions: button, knob, slider and toggle. Input method had two conditions: controller-based input (CTRL) and hand-tracked input (HAND). Haptic feedback had two conditions: passive haptic feedback (HAP) and no haptic feedback (NOH). In HAP, the vertical surface of the virtual wall was aligned with a real-world wall surface. Participants used each input method to interact with each control

*e-mail: markus.tatzgern@fh-salzburg.ac.at)

†e-mail: david.fasching@fh-salzburg.ac.at

‡e-mail: vn.gomez@uniandes.edu.co

§e-mail: s.calvache@uniandes.edu.co

¶e-mail: pfiguero@uniandes.edu.co

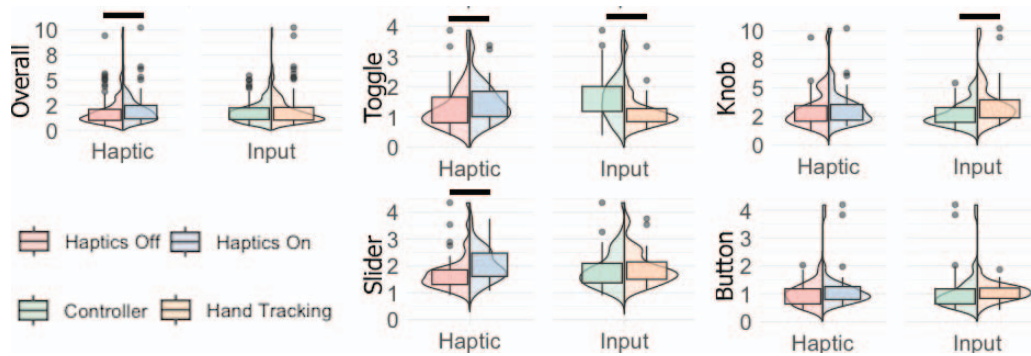


Figure 2: Violin plots and box plots for task completion time for each control element and factors haptic feedback and input method. Statistically significant differences between input methods and haptic feedback conditions are indicated as lines.

element, with and without haptic feedback. 20 participants (6 female, $\bar{X}=23.6$ (4.2) years) volunteered from two universities.

Hypotheses. **H1.** Haptic feedback HAP would improve the performance of interactions with control elements in general, as it allows users to be more precise and react faster to touching elements. **H2.** Controller-based input CTRL would be objectively faster than hand input HAND as it allows for precise interactions. Hand tracking would not be sufficiently precise for interacting with small-sized control elements as utilized in this study. **H3.** Hand input HAND would outperform controller input CTRL in subjective measures.

Results. For task completion time (TCT) see Fig. 2. **Haptic Feedback decreases Interaction Performance.** Surprisingly, haptic feedback appears to have slowed down participants, as indicated by the statistically significant difference in TCT which contradicts previous research [2, 8]. We found this effect for toggle and slider element. There are several possible reasons for this outcome. First, there is a mismatch between the touchpoints of a physical control element and its abstraction to a flat wall surface, which may make the haptic feedback less natural. Second, another reason may be the offset between the control element and the wall surface. Participants calibrated the wall to be aligned with the real wall. Hence, control elements were reached out of the physical wall. This offset was required to enable participants to interact with elements, but also allowed them to interact without touching the wall. Participants did not remark on this offset being a conscious performance issue during interaction in the interviews. Third, 35% of participants themselves remarked on the fear of hitting the wall as a major issue during interaction.

Hand Tracking Preference. The precisely tracked controller input did not outperform hand tracking. Hand tracking even outperformed controller input for the toggle element, where both input methods had similarly high fidelity. However, participants generally preferred the high fidelity hand input method over the controller. This indicates that state-of-the-art off-the-shelf hand-tracked input may be a feasible method to interact with VR control elements. However, vision-based hand tracking still has limitations, such as a lack of fine motor control and the limited field of view of tracking cameras, that we have not extensively explored in our experiment [1].

Fine Motor Manipulation. The knob element showed the limitations of hand input using vision-based tracking. 35% of participants noted that interacting with the knob using the hand input was unnatural. Participants expected to be able to twist the knob using their thumb and index finger, instead of being required to rotate the whole forearm. Interestingly, previous work [11] determined that a lower fidelity hand input approximation using capacitive sensing of controllers that closely resembles real hand interaction was preferred by users over a standard raycasting input method. Our natural hand input method differs from the hand input approximation by not us-

ing a controller for capturing hand gestures. Hence, the lack of fine motor input in our case may have moved the input method towards the uncanny valley of interaction fidelity [10], where an otherwise natural interaction does not behave as expected. Fine motor manipulations may be enabled by capturing multiple viewpoints, adding additional markers to fingers, or utilizing a glove or exoskeleton [5]. However, these methods would likely add more hardware to an otherwise standalone VR head-mounted display (HMD) experience.

ACKNOWLEDGMENTS

This work was supported by a grant from the Federal State Government of Salzburg (WISS 2025 grant no. 20102-F2002459-KZP).

REFERENCES

- [1] S. Auer, J. Gerken, H. Reiterer, and H.-C. Jetter. Comparison between virtual reality and physical flight simulators for cockpit familiarization. In *Proceedings of Mensch Und Computer 2021*, MuC '21, p. 378–392, 2021. doi: 10.1145/3473856.3473860
- [2] C. W. Borst and R. A. Volz. Evaluation of a haptic mixed reality system for interactions with a virtual control panel. *Presence: Teleoperators & Virtual Environments*, 14(6):677–696, 2005.
- [3] L.-P. Cheng, E. Ofek, C. Holz, H. Benko, and A. D. Wilson. Sparse haptic proxy: Touch feedback in virtual environments using a general passive prop. In *2017 CHI Conf. on Human Factors in Computing Systems*, CHI '17, p. 3718–3728, 2017. doi: 10.1145/3025453.3025753
- [4] C. Cho, H. Yang, G. Kim, and S. Han. Body-based interfaces. In *Proceedings. Fourth IEEE International Conference on Multimodal Interfaces*, pp. 466–472, 2002. doi: 10.1109/ICMI.2002.1167040
- [5] S. Jörg, Y. Ye, M. Neff, F. Mueller, and V. Zordan. Virtual hands in vr: motion capture, synthesis, and perception. In *ACM SIGGRAPH 2020 Courses*, pp. 1–145. ACM, 2020.
- [6] R. D. Joyce and S. Robinson. *Passive Haptics to Enhance Virtual Reality Simulations*. 2017. doi: 10.2514/6.2017-1313
- [7] L. Kohli and M. Whitton. The haptic hand: providing user interface feedback with the non-dominant hand in virtual environments. In *Proceedings of Graphics Interface 2005*, pp. 1–8, 2005.
- [8] R. W. Lindeman, J. L. Sibert, and J. K. Hahn. Hand-held windows: towards effective 2d interaction in immersive virtual environments. In *IEEE VR*, pp. 205–212, 1999.
- [9] R. W. Lindeman, J. L. Sibert, and J. N. Templeman. The effect of 3d widget representation and simulated surface constraints on interaction in virtual environments. In *IEEE VR 2001*, pp. 141–148, 2001.
- [10] R. P. McMahan, C. Lai, and S. K. Pal. Interaction fidelity: the uncanny valley of virtual reality interactions. In *International Conference on Virtual, Augmented and Mixed Reality*, pp. 59–70. Springer, 2016.
- [11] M. Tatzgern and C. Birgmann. Exploring input approximations for control panels in virtual reality. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*, pp. 1–9, 2021. doi: 10.1109/VR50410.2021.00092